Cybersecurity - Undergraduate Level Assessment 16 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. Let I = (S,V)I=(S,V) be a MAC. Suppose S(k,m)S(k,m) is always 5 bits long. Can this MAC be secure?
   (a) No, an attacker can simply guess the tag for messages
   (b) It depends on the details of the MAC
   (c) Yes, the attacker cannot generate a valid tag for any message
   (d) Yes, the PRG is pseudorandom
2. A sender must not be able to deny sending a message that was sent, is known as
   (a) Message Nonrepudiation
   (b) Message Integrity
   (c) Message Condentiality
   (d) Message Sending
3. Which Nmap scan is does not completely open a TCP connection?
   (a) SYN stealth scan
   (b) TCP connect
   (c) XMAS tree scan
   (d) ACK scan
4. What are the port states determined by Nmap?
   (a) Active, inactive, standby
   (b) Open, half-open, closed
   (c) Open, filtered, unfiltered
   (d) Active, closed, unused
5. __________________ is also a part of darknet that is employed for transferring files anonymously.
   (a) Freenet
   (b) ARPANET
   (c) Stuxnet
   (d) Internet

True / False (10 marks)

Answer True or False

6. True or False: WPA 3 offers the strongest wireless security features compared to previous standards like WEP, WPA, and WPA 2.
7. True or False: WiFi traffic sniffing specifically targets only encrypted wireless networks for analysis and does not apply to unencrypted traffic.
8. True or False: Authorization is primarily concerned with identifying whether a user is an attacker or not.
9. True or False: The four primary security principles related to messages are Confidentiality, Integrity, Non-repudiation, and Authentication.
10. True or False: Message authentication ensures the integrity of a message by verifying its origin and preventing tampering.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a function to find the perimeter of a pentagon.
12. Write a function to find all pairs in an integer array whose sum is equal to a given number.

End of Paper